

## Old UNM Algebra Qualifier Problems

Dummit–Foote, Chapter 3

Problems with Detailed Solutions

This document supplies a solution immediately after each problem in the deduplicated Chapter 3 UNM packet. The statements have been preserved apart from minor typographical normalization. Problems already included in the earlier Chapters 1–2 Qual packet remain excluded. In Section II, the solutions explicitly identify later results such as Sylow theory, the class equation, or Burnside’s theorem when they are used.

### I. Core problems from Chapter 3

#### January 2013 Qualifier, Problem 2

Suppose that  $N$  is a normal subgroup of a group  $G$ , with  $|N| < \infty$ . Suppose also that  $H$  is a subgroup of  $G$  with  $[G : H] < \infty$  and that  $|N|$  and  $[G : H]$  are relatively prime. Show that  $N \leq H$ .

**Solution.** Because  $N \trianglelefteq G$ , the product  $NH$  is a subgroup of  $G$ . The second isomorphism theorem gives

$$NH/H \cong N/(N \cap H),$$

as a statement about coset spaces (and as a group isomorphism when  $H$  is normal in  $NH$ ). In particular,

$$[NH : H] = [N : N \cap H].$$

The right-hand side divides  $|N|$  by Lagrange’s theorem. The left-hand side divides  $[G : H]$ , since

$$[G : H] = [G : NH][NH : H].$$

Thus  $[N : N \cap H]$  divides two relatively prime integers and must equal 1. Consequently  $N = N \cap H$ , so  $N \leq H$ .

#### January 2013 Qualifier, Problem 4

Let  $F$  be the free group on two generators and let  $[F, F]$  be its commutator subgroup. Prove that

$$F/[F, F] \cong \mathbb{Z} \times \mathbb{Z}.$$

**Solution.** Let  $x, y$  be the free generators of  $F$ . By the universal property of the free group, the assignments

$$x \mapsto (1, 0), \quad y \mapsto (0, 1)$$

extend uniquely to a homomorphism

$$\varepsilon : F \longrightarrow \mathbb{Z}^2.$$

The map is surjective. Since  $\mathbb{Z}^2$  is abelian, every commutator lies in its kernel, so  $[F, F] \leq \ker \varepsilon$ .

It remains to prove the reverse inclusion. In the quotient  $F/[F, F]$ , the images of  $x$  and  $y$  commute. Therefore the image of any word  $w$  is

$$w[F, F] = x^m y^n [F, F],$$

where  $m$  and  $n$  are the total exponent sums of  $x$  and  $y$  in  $w$ . If  $w \in \ker \varepsilon$ , then  $m = n = 0$ , hence  $w[F, F] = [F, F]$ . Thus  $w \in [F, F]$ , proving  $\ker \varepsilon = [F, F]$ . The first isomorphism theorem now yields

$$F/[F, F] \cong \mathbb{Z}^2.$$

### August 2013 Qualifier, Problem 1

Prove that if  $H$  and  $K$  are finite subgroups of  $G$  whose orders are relatively prime, then  $H \cap K = \{e\}$ .

**Solution.** The subgroup  $H \cap K$  is contained in both  $H$  and  $K$ . Hence Lagrange's theorem implies that  $|H \cap K|$  divides both  $|H|$  and  $|K|$ . Since these two orders are relatively prime,  $|H \cap K| = 1$ . Therefore  $H \cap K = \{e\}$ .

### August 2013 Qualifier, Problem 2

Let  $G$  be a group and let  $H$  be a subgroup contained in the center of  $G$ . Show that if  $G/H$  is cyclic, then  $G$  is abelian.

**Solution.** Choose  $g \in G$  such that  $G/H = \langle gH \rangle$ . For arbitrary  $x, y \in G$ , there exist integers  $m, n$  and elements  $h_1, h_2 \in H$  such that

$$x = g^m h_1, \text{ and } y = g^n h_2.$$

Because  $H \leq Z(G)$ , the elements  $h_1, h_2$  commute with every element of  $G$ . Therefore

$$xy = g^{m+n} h_1 h_2 = g^{n+m} h_2 h_1 = yx.$$

Thus every two elements of  $G$  commute, so  $G$  is abelian.

### January 2015 Qualifier, Problem 9

Prove that the group  $S_4$  is solvable.

**Solution.** Let

$$V = \{e, (12)(34), (13)(24), (14)(23)\}.$$

This is the Klein four subgroup of  $S_4$ . Conjugation preserves cycle type, so  $V \trianglelefteq S_4$ . In fact  $V \leq A_4$ , and the chain

$$\{e\} \trianglelefteq V \trianglelefteq A_4 \trianglelefteq S_4$$

is a normal series. Its factors are

$$V \cong C_2 \times C_2, \text{ and } A_4/V \cong C_3, \quad S_4/A_4 \cong C_2.$$

All three factors are abelian. Hence this is a normal series with abelian factors, and  $S_4$  is solvable.

**August 2016 Qualifier, Problem 3**

Prove that the group of invertible upper-triangular  $n \times n$  complex matrices is solvable.

**Solution.** Let  $B$  denote this group and let  $U$  be its subgroup of upper-unitriangular matrices, namely matrices whose diagonal entries are all 1. The map

$$B \longrightarrow (\mathbb{C}^\times)^n, \text{quadr}(a_{ij}) \longmapsto (a_{11}, \dots, a_{nn})$$

is a surjective homomorphism with kernel  $U$ . Thus  $U \trianglelefteq B$  and  $B/U$  is abelian.

For  $r \geq 1$ , let  $U_r$  consist of those matrices in  $U$  whose entries on the first  $r - 1$  superdiagonals are zero. Thus  $U_1 = U$  and  $U_n = \{I\}$ . A direct multiplication calculation gives

$$[U_r, U_s] \subseteq U_{r+s}.$$

Consequently the derived series satisfies

$$U^{(j)} \subseteq U_{2^j}.$$

For  $2^j \geq n$  this subgroup is trivial, so  $U$  is solvable. Since  $U$  is solvable and  $B/U$  is abelian,  $B$  is solvable.

**August 2017 Qualifier, Problem 1**

Suppose  $N$  and  $H$  are normal subgroups of  $G$  such that  $G/N$  and  $G/H$  are abelian. Show that  $G/(H \cap N)$  is abelian.

**Solution.** Since  $G/N$  is abelian, every commutator of elements of  $G$  lies in  $N$ ; equivalently,  $[G, G] \leq N$ . Similarly, the abelianness of  $G/H$  gives  $[G, G] \leq H$ . Hence

$$[G, G] \leq H \cap N.$$

The subgroup  $H \cap N$  is normal in  $G$ . A quotient  $G/K$  is abelian exactly when  $[G, G] \leq K$ . Applying this criterion to  $K = H \cap N$  shows that  $G/(H \cap N)$  is abelian.

**August 2018 Qualifier, Problem 2**

Let  $N$  be a normal subgroup of a group  $G$ . Show that  $N$  contains the commutator subgroup if and only if  $G/N$  is abelian.

**Solution.** Let  $\pi : G \rightarrow G/N$  be the quotient map. For  $x, y \in G$ ,

$$[\pi(x), \pi(y)] = \pi([x, y]).$$

If  $[G, G] \leq N$ , then  $\pi([x, y]) = N$  for all  $x, y$ , so all elements of  $G/N$  commute and  $G/N$  is abelian.

Conversely, if  $G/N$  is abelian, then  $[\pi(x), \pi(y)] = N$  for every  $x, y \in G$ . Hence  $[x, y] \in \ker \pi = N$ . Since  $[G, G]$  is generated by all such commutators,  $[G, G] \leq N$ .

**August 2018 Qualifier, Problem 7**

Prove that the group of invertible upper-triangular  $n \times n$  matrices with coefficients in a field is solvable.

**Solution.** Let  $F$  be the coefficient field, let  $B$  be the upper-triangular group, and let  $U$  be the upper-unitriangular subgroup. Taking diagonal entries gives an exact sequence

$$1 \longrightarrow U \longrightarrow B \longrightarrow (F^\times)^n \longrightarrow 1.$$

Thus  $B/U$  is abelian. Define  $U_r$  by requiring the first  $r - 1$  superdiagonals to vanish. Matrix multiplication gives

$$[U_r, U_s] \subseteq U_{r+s},$$

and therefore  $U^{(j)} \subseteq U_{2j}$ . Since  $U_n = 1$ , the derived series of  $U$  terminates. Hence  $U$  is solvable; extending it by the abelian group  $B/U$  shows that  $B$  is solvable.

**January 2020 Qualifier, Problem 2**

Suppose  $H$  and  $N$  are subgroups of a group  $G$ , with  $N \trianglelefteq G$ ,  $|H| = m < \infty$ , and  $[G : N] = n < \infty$ . If  $m$  and  $n$  are relatively prime, prove that  $H \leq N$ .

**Solution.** Let  $\pi : G \rightarrow G/N$  be the quotient map. The restriction  $\pi|_H$  has kernel  $H \cap N$ , so

$$|\pi(H)| = [H : H \cap N].$$

This order divides  $|H| = m$ . Since  $\pi(H)$  is a subgroup of the finite group  $G/N$ , it also divides  $|G/N| = [G : N] = n$ . The integers  $m$  and  $n$  are coprime, hence  $|\pi(H)| = 1$ . Therefore every element of  $H$  lies in  $\ker \pi = N$ , and  $H \leq N$ .

**January 2021 Qualifier, Problem 2**

Prove that the group of invertible  $2 \times 2$  complex matrices is not solvable.

**Solution.** It suffices to exhibit a nonsolvable subgroup of  $\mathrm{GL}_2(\mathbb{C})$ . We show that  $\mathrm{SL}_2(\mathbb{C})$  is perfect. Put

$$u(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \ell(t) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}, d(a) = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

The elementary matrices  $u(t)$  and  $\ell(t)$  generate  $\mathrm{SL}_2(\mathbb{C})$  by Gaussian elimination. For completeness, diagonal matrices also lie in the subgroup they generate because, with

$$w(a) = u(a)\ell(-a^{-1})u(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix},$$

one has  $d(a) = w(a)w(-1)$ .

Choose  $a \in \mathbb{C}^\times$  with  $a^2 \neq 1$ . Direct calculation gives

$$[d(a), u(s)] = u((a^2 - 1)s).$$

As  $s$  varies,  $(a^2 - 1)s$  runs through all of  $\mathbb{C}$ , so every  $u(t)$  is a commutator. Similarly, every  $\ell(t)$  is a commutator. Since these matrices generate  $\mathrm{SL}_2(\mathbb{C})$ ,

$$[\mathrm{SL}_2(\mathbb{C}), \mathrm{SL}_2(\mathbb{C})] = \mathrm{SL}_2(\mathbb{C}).$$

Thus its derived series never reaches the identity, so it is not solvable. Every subgroup of a solvable group is solvable; therefore  $\mathrm{GL}_2(\mathbb{C})$  cannot be solvable.

**January 2023 Qualifier, Problem 1**

Suppose  $N$  is a normal subgroup of a finite group  $G$  and

$$\gcd(|N|, (G : N)) = 1.$$

Show that  $N$  is the unique subgroup of  $G$  of order  $|N|$ .

**Solution.** Let  $H \leq G$  have  $|H| = |N|$ . Under the quotient map  $\pi : G \rightarrow G/N$ , the image  $\pi(H)$  has order

$$|\pi(H)| = [H : H \cap N].$$

This divides  $|H| = |N|$  and also divides  $|G/N| = (G : N)$ . The two latter integers are coprime, so  $|\pi(H)| = 1$ . Hence  $H \leq N$ . Since  $H$  and  $N$  are finite and have the same order,  $H = N$ . Thus  $N$  is the unique subgroup of  $G$  of order  $|N|$ .

**August 2023 Qualifier, Problem 3**

Prove that the group of invertible upper-triangular matrices with real coefficients is solvable.

**Solution.** For  $n \times n$  matrices, let  $B$  be the upper-triangular group and  $U$  its upper-unitriangular subgroup. The diagonal map has kernel  $U$  and abelian image  $(\mathbb{R}^\times)^n$ , so  $B/U$  is abelian. If  $U_r$  denotes the subgroup with the first  $r - 1$  superdiagonals zero, then

$$[U_r, U_s] \subseteq U_{r+s}.$$

It follows inductively that  $U^{(j)} \subseteq U_{2j}$ . Since  $U_n = 1$ , the group  $U$  is solvable. An extension of a solvable group by an abelian group is solvable, hence  $B$  is solvable.

**August 2024 Qualifier, Problem 1**

Suppose

$$1 \longrightarrow H \xrightarrow{\alpha} G \xrightarrow{\beta} K \longrightarrow 1$$

is an exact sequence of groups such that  $H$  and  $K$  are abelian and  $\alpha(H)$  is central in  $G$ . Let  $a$  be a fixed element of  $G$ . Show that

$$\varphi : G \longrightarrow G, \quad \varphi(g) = aga^{-1}g^{-1},$$

is a group homomorphism.

**Solution.** Because  $K$  is abelian,

$$\beta(aga^{-1}g^{-1}) = [\beta(a), \beta(g)] = 1.$$

Exactness therefore gives

$$\varphi(g) \in \ker \beta = \alpha(H) \subseteq Z(G)$$

for every  $g \in G$ . For  $g, h \in G$ , the standard commutator identity gives

$$\begin{aligned} \varphi(gh) &= a(gh)a^{-1}(gh)^{-1} \\ &= (aga^{-1}g^{-1})g(aha^{-1}h^{-1})g^{-1} \\ &= \varphi(g)g\varphi(h)g^{-1}. \end{aligned}$$

But  $\varphi(h)$  is central, so  $g\varphi(h)g^{-1} = \varphi(h)$ . Hence

$$\varphi(gh) = \varphi(g)\varphi(h),$$

and  $\varphi$  is a homomorphism.

## II. Closely related or later-material problems

### January 2012 Qualifier, Problem 2

Prove that every group of order  $pq$ , where  $p$  and  $q$  are unequal primes, is solvable.

**Solution.** Assume  $p < q$ . By Sylow's theorems, the number  $n_q$  of Sylow  $q$ -subgroups satisfies

$$n_q \mid p, \text{ and } n_q \equiv 1 \pmod{q}.$$

The only possibility is  $n_q = 1$ . Hence the Sylow  $q$ -subgroup  $Q$  is normal in  $G$ . Both  $Q$  and  $G/Q$  have prime order and are therefore cyclic. Thus

$$1 \trianglelefteq Q \trianglelefteq G$$

is a normal series with abelian factors, so  $G$  is solvable.

### August 2012 Qualifier, Problem 1

Let  $p, q, r$  be primes. Show that no group of order  $pqr$  is simple.

**Solution.** If two of the primes coincide, then  $|G|$  is divisible by only two distinct primes. Burnside's  $p^a q^b$  theorem makes  $G$  solvable. A nontrivial solvable simple group has prime order, whereas  $pqr$  is composite; hence  $G$  is not simple.

Now suppose  $p < q < r$  are distinct and, toward a contradiction, that  $G$  is simple. The number  $n_r$  of Sylow  $r$ -subgroups divides  $pq$  and is congruent to 1 modulo  $r$ . Since it is not 1, necessarily  $n_r = pq$ . Distinct Sylow  $r$ -subgroups intersect trivially, so they contain

$$pq(r-1) = pqr - pq$$

nonidentity elements.

Similarly,  $n_q$  divides  $pr$  and is 1 modulo  $q$ . Simplicity rules out  $n_q = 1$ , and  $n_q \neq p$  because  $p < q$ . Thus  $n_q$  is either  $r$  or  $pr$ . If  $n_q = pr$ , the Sylow  $q$ -subgroups contribute  $pr(q-1)$  further nonidentity elements, and

$$pq(r-1) + pr(q-1) > pqr - 1,$$

which is impossible. If  $n_q = r$ , they contribute  $r(q-1)$  elements, and again

$$pq(r-1) + r(q-1) > pqr - 1.$$

Indeed the difference in the second inequality is  $(q-1)r - pq + 1 > 0$  because  $r > q > p$ . Both possibilities are impossible. Therefore  $G$  cannot be simple.

### January 2013 Qualifier, Problem 3

Show that there are no simple groups of order 196.

**Solution.** Since  $196 = 2^2 \cdot 7^2$ , the number  $n_7$  of Sylow 7-subgroups satisfies

$$n_7 \mid 4, \text{ and } n_7 \equiv 1 \pmod{7}.$$

Thus  $n_7 = 1$ . The unique Sylow 7-subgroup is normal and nontrivial, so a group of order 196 is not simple.

**August 2013 Qualifier, Problem 3**

Show that no group of order 132 is simple.

**Solution.** Write  $132 = 2^2 \cdot 3 \cdot 11$ . Suppose  $G$  were simple. The number of Sylow 11-subgroups satisfies

$$n_{11} \mid 12, \text{quad} n_{11} \equiv 1 \pmod{11}.$$

Thus  $n_{11} = 12$ ; otherwise the Sylow 11-subgroup would be normal. These subgroups contain  $12(11 - 1) = 120$  nonidentity elements.

Next,  $n_3 \mid 44$  and  $n_3 \equiv 1 \pmod{3}$ , so  $n_3 \in \{1, 4, 22\}$ . Simplicity excludes 1, while 22 Sylow 3-subgroups would contribute  $22(3 - 1) = 44$  elements, more than the eleven nonidentity elements not already lying in Sylow 11-subgroups. Hence  $n_3 = 4$ .

Conjugation on the four Sylow 3-subgroups gives a nontrivial homomorphism  $G \rightarrow S_4$ . Its kernel is normal. If  $G$  were simple, the kernel would be trivial, because the action is nontrivial. This would embed a group of order 132 in  $S_4$ , which has order 24, an impossibility. Therefore  $G$  is not simple.

**January 2014 Qualifier, Problem 3**

Let  $G$  be a group of order 105 with a normal Sylow 3-subgroup. Show that  $G$  is abelian.

**Solution.** Let  $P_3$  be the normal Sylow 3-subgroup. Sylow's theorems give

$$n_7 \in \{1, 15\}, \text{quad} n_5 \in \{1, 6, 21\}.$$

If  $n_7 = 15$ , the Sylow 7-subgroups contain  $15(7 - 1) = 90$  nonidentity elements. If at the same time  $n_5 > 1$ , the Sylow 5-subgroups contain at least  $6(5 - 1) = 24$  further nonidentity elements. Together with the two nonidentity elements of  $P_3$ , this exceeds 104. Thus  $n_5 = 1$  whenever  $n_7 = 15$ . Consequently, besides  $P_3$ , at least one of a Sylow 5-subgroup or a Sylow 7-subgroup is normal.

Let  $P_q$  be such a normal subgroup, where  $q \in \{5, 7\}$ , and put  $N = P_3 P_q$ . Normal subgroups of relatively prime orders commute, because  $[P_3, P_q] \leq P_3 \cap P_q = 1$ . Hence

$$N \cong C_3 \times C_q \cong C_{3q}.$$

Let  $P_s$  be a Sylow subgroup for the remaining prime  $s$ . Then  $G = NP_s$ . Conjugation defines a homomorphism

$$P_s \longrightarrow \text{Aut}(N).$$

If  $N \cong C_{15}$ , then  $|\text{Aut}(N)| = \varphi(15) = 8$ , which is not divisible by 7; if  $N \cong C_{21}$ , then  $|\text{Aut}(N)| = \varphi(21) = 12$ , which is not divisible by 5. Thus the homomorphism is trivial in either case. Therefore  $P_s$  centralizes  $N$ , and

$$G = N \times P_s$$

is abelian (indeed cyclic of order 105).

**January 2015 Qualifier, Problem 1**

Let  $G$  be a nonabelian group of order  $p^3$ . Show that

$$Z(G) = G' = [G, G].$$

**Solution.** The class equation for a finite  $p$ -group implies  $Z(G) \neq 1$ . If  $|Z(G)| \geq p^2$ , then  $|G/Z(G)| \leq p$ , so  $G/Z(G)$  is cyclic. A cyclic quotient by the center forces  $G$  to be abelian, contrary to hypothesis. Hence  $|Z(G)| = p$ .

The quotient  $G/Z(G)$  has order  $p^2$  and is therefore abelian. Thus  $[G, G] \leq Z(G)$ . Since  $G$  is nonabelian,  $[G, G] \neq 1$ . But  $Z(G)$  has prime order, so its only nontrivial subgroup is itself. Therefore

$$[G, G] = Z(G).$$

**January 2015 Qualifier, Problem 2**

Show that there is no simple group of order 192.

**Solution.** Since  $192 = 2^6 \cdot 3$ , the number  $n_2$  of Sylow 2-subgroups divides 3 and is odd. Hence  $n_2 = 1$  or 3. In a simple group it would have to be 3. Conjugation on the three Sylow 2-subgroups would then give a nontrivial homomorphism

$$G \longrightarrow S_3.$$

The kernel is normal. Simplicity would force the kernel to be trivial, since the action is nontrivial, embedding a group of order 192 into a group of order 6. This is impossible. Hence no group of order 192 is simple.

**August 2017 Qualifier, Problem 2**

Show that every group of order 100 is not simple.

**Solution.** We have  $100 = 2^2 \cdot 5^2$ . The number  $n_5$  of Sylow 5-subgroups divides 4 and satisfies  $n_5 \equiv 1 \pmod{5}$ . Thus  $n_5 = 1$ , so the Sylow 5-subgroup is normal and the group is not simple.

**January 2018 Qualifier, Problem 6**

Let  $p$  be a prime. Show that for every nonabelian group  $G$  of order  $p^3$ , the commutator subgroup of  $G$  is equal to the center of  $G$ .

**Solution.** By the class equation, the center of a finite  $p$ -group is nontrivial. If  $|Z(G)|$  were  $p^2$  or  $p^3$ , then  $G/Z(G)$  would be cyclic, forcing  $G$  to be abelian. Hence  $|Z(G)| = p$ . Since  $G/Z(G)$  has order  $p^2$ , it is abelian, and therefore  $[G, G] \leq Z(G)$ . Nonabelianness gives  $[G, G] \neq 1$ . The group  $Z(G)$  has prime order, so  $[G, G] = Z(G)$ .

**August 2018 Qualifier, Problem 1**

Show that every group of order  $3^3 \cdot 13$  is not simple.

**Solution.** Let  $G$  have order  $3^3 \cdot 13$ . The number of Sylow 13-subgroups satisfies

$$n_{13} \mid 27, \text{ and } n_{13} \equiv 1 \pmod{13}.$$

Thus  $n_{13} = 1$  or  $27$ . In the first case the Sylow 13-subgroup is normal. In the second case, the Sylow 13-subgroups contain

$$27(13 - 1) = 324$$

nonidentity elements. Exactly 27 elements remain, including the identity. Any Sylow 3-subgroup has order 27, so it is precisely this remaining set. It is therefore the unique Sylow 3-subgroup and is normal. In either case  $G$  has a nontrivial proper normal subgroup and is not simple.

**January 2019 Qualifier, Problem 2**

Show that no group of order 52 is simple.

**Solution.** Since  $52 = 2^2 \cdot 13$ , the number  $n_{13}$  of Sylow 13-subgroups divides 4 and is congruent to 1 modulo 13. Hence  $n_{13} = 1$ . The unique Sylow 13-subgroup is normal, so the group is not simple.

**August 2019 Qualifier, Problem 3**

Let  $K$  be a field and let  $G$  be the set of all functions  $f : K \rightarrow K$  of the form

$$f(x) = ax + b,$$

where  $a, b \in K$  and  $a \neq 0$ . View  $G$  as a group under composition. Prove that  $G$  is solvable. Is  $G$  nilpotent? Justify your answer.

**Solution.** Write  $t_b(x) = x + b$  and  $d_a(x) = ax$ . The translations

$$T = \{t_b : b \in K\}$$

form a normal abelian subgroup of  $G$ , isomorphic to the additive group  $(K, +)$ . The homomorphism

$$G \longrightarrow K^\times, \text{ and } (x \mapsto ax + b) \longmapsto a$$

is surjective with kernel  $T$ . Since  $K^\times$  is abelian, both  $T$  and  $G/T$  are abelian. Thus  $G$  is solvable of derived length at most 2.

For nilpotence, compute

$$[d_a, t_b] = t_{(a-1)b}.$$

If  $|K| > 2$ , choose  $a \neq 0, 1$ . Since  $a - 1 \neq 0$ , the elements  $(a - 1)b$  run through all of  $K$  as  $b$  varies. Hence

$$[G, G] = T \quad \text{and} \quad [T, G] = T.$$

The lower central series therefore stabilizes at the nontrivial subgroup  $T$ , so  $G$  is not nilpotent. If  $K = \mathbb{F}_2$ , then  $K^\times = \{1\}$  and  $G = T \cong C_2$ , which is nilpotent. Thus  $G$  is nilpotent exactly when  $K = \mathbb{F}_2$ .

**January 2020 Qualifier, Problem 1**

Let  $G$  be a group of order 402. Prove that  $G$  is not simple.

**Solution.** Since  $402 = 2 \cdot 3 \cdot 67$ , the number  $n_{67}$  of Sylow 67-subgroups divides 6 and satisfies  $n_{67} \equiv 1 \pmod{67}$ . The only possibility is  $n_{67} = 1$ . Thus the Sylow 67-subgroup is normal and  $G$  is not simple.

**August 2021 Qualifier, Problem 1**

Show that no group of order 72 is simple.

**Solution.** We have  $72 = 2^3 \cdot 3^2$ . The number  $n_3$  of Sylow 3-subgroups divides 8 and is congruent to 1 modulo 3. Hence  $n_3 = 1$  or 4. If  $n_3 = 1$ , the Sylow 3-subgroup is normal. If  $n_3 = 4$ , conjugation on the four Sylow 3-subgroups gives a nontrivial homomorphism  $G \rightarrow S_4$ . Were  $G$  simple, the kernel would be trivial, embedding a group of order 72 in  $S_4$ , which has order 24. This is impossible. Hence  $G$  is not simple.

**January 2022 Qualifier, Problem 1**

Show that no group of order 132 is simple.

**Solution.** Suppose  $|G| = 132 = 2^2 \cdot 3 \cdot 11$  and  $G$  is simple. Then  $n_{11} = 12$ , so the Sylow 11-subgroups contain 120 nonidentity elements. Also

$$n_3 \mid 44, \text{ and } n_3 \equiv 1 \pmod{3},$$

so  $n_3 \in \{1, 4, 22\}$ . Simplicity excludes 1, and 22 would require 44 additional nonidentity elements, although only eleven remain. Thus  $n_3 = 4$ . The conjugation action on the four Sylow 3-subgroups is nontrivial and gives  $G \rightarrow S_4$ . Simplicity forces this map to be injective, which is impossible because  $132 > 24$ . Therefore  $G$  is not simple.

**January 2022 Qualifier, Problem 2**

Let  $p$  be an odd prime and let  $G$  be a group of order  $p^3$ . Show that

$$Z(G) = [G, G].$$

**Correction to the printed problem.** As written, the assertion is false: for example, if  $G = C_{p^3}$ , then  $Z(G) = G$  while  $[G, G] = 1$ . The intended statement requires  $G$  to be nonabelian.

**Solution.** Under the intended nonabelian hypothesis, the class equation gives  $Z(G) \neq 1$ . If  $|Z(G)| \geq p^2$ , then  $G/Z(G)$  is cyclic and  $G$  is abelian, a contradiction. Hence  $|Z(G)| = p$ . The quotient  $G/Z(G)$  has order  $p^2$  and is abelian, so  $[G, G] \leq Z(G)$ . Since  $G$  is nonabelian,  $[G, G] \neq 1$ ; because  $Z(G)$  has prime order,  $[G, G] = Z(G)$ .

**January 2024 Qualifier, Problem 1**

Show that there is no simple group of order 80.

**Solution.** Since  $80 = 2^4 \cdot 5$ , the number of Sylow 5-subgroups satisfies

$$n_5 \mid 16, n_5 \equiv 1 \pmod{5}.$$

Thus  $n_5 = 1$  or 16. In the first case the Sylow 5-subgroup is normal. If  $n_5 = 16$ , these subgroups contain  $16(5 - 1) = 64$  nonidentity elements. The remaining set has 16 elements including the identity. Every Sylow 2-subgroup has order 16 and must equal this remaining set. Hence it is the unique Sylow 2-subgroup and is normal. In either case the group is not simple.

**August 2024 Qualifier, Problem 2**

Let  $G$  be a group of order  $3 \cdot 2^n$ , where  $n \geq 1$ . Show that  $G$  is solvable.

**Solution.** Burnside's  $p^a q^b$  theorem states that every finite group whose order is divisible by at most two distinct primes is solvable. Here

$$|G| = 2^n 3^1,$$

so the theorem applies directly and shows that  $G$  is solvable.

The reliance on Burnside's theorem is substantive: for arbitrary  $n$ , a short Sylow count alone does not always produce an immediately visible normal Sylow subgroup. This is why the problem belongs in the later-material section rather than the core Chapter 3 list.

**Source**

University of New Mexico, Department of Mathematics and Statistics, **Past Qualifying Exams – Algebra**. The group-theory terminology and standard results follow Dummit and Foote, *Abstract Algebra*, 3rd ed., especially Chapters 3 and 4.

Each solution is placed immediately after its corresponding archived problem.